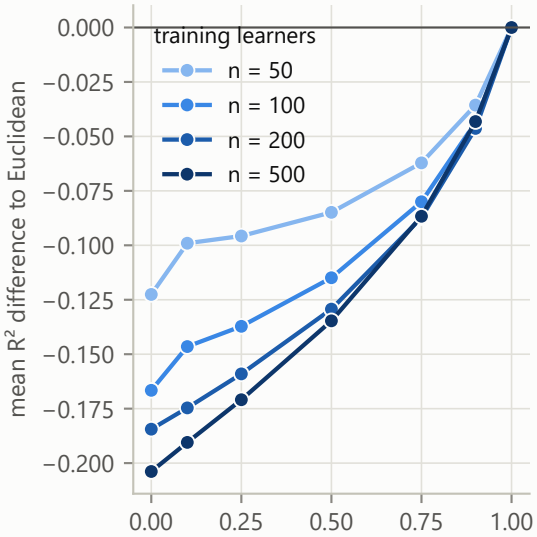
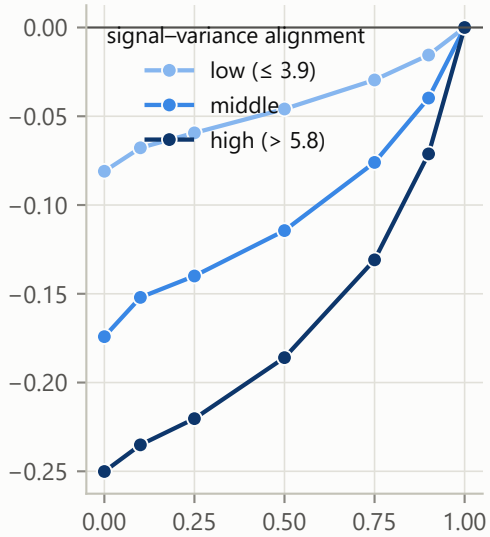
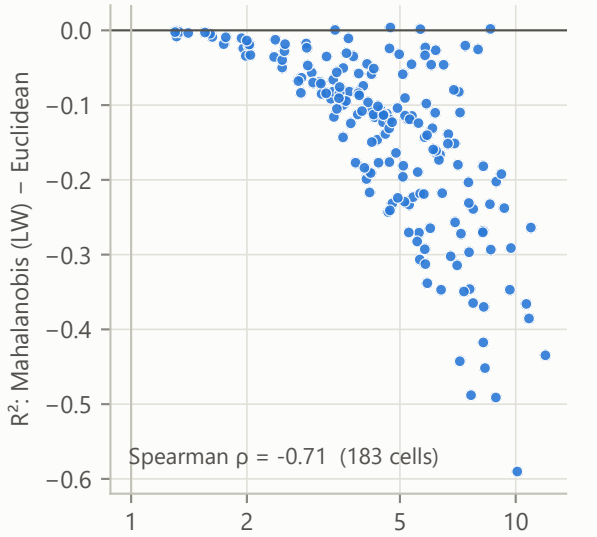

(a) Path from Mahalanobis to Euclidean
training learners
n = 50
n = 100
n = 200
n = 500
mean R² difference to Euclidean
0.000
−0.025
−0.050
−0.075
−0.100
−0.125
−0.150
−0.175
−0.200
0.00
0.25
0.50
0.75
1.00
shrinkage weight α on the identity
(0 = sample Mahalanobis, 1 = Euclidean)


(b) The same path, by alignment tercile
signal–variance alignment
low (≤ 3.9)
middle
high (> 5.8)
0.00
−0.05
−0.10
−0.15
−0.20
−0.25
0.00
0.25
0.50
0.75
1.00
shrinkage weight α on the identity


(c) Whitening effect against alignment, per cell
R²: Mahalanobis (LW) − Euclidean
0.0
−0.1
−0.2
−0.3
−0.4
−0.5
−0.6
Spearman ρ = -0.71 (183 cells)
1
2
5
10
signal–variance alignment (log scale)
> 1: outcome carried by high-variance directions